MAE 3 Pendulum Clock Project

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https://youtube.com/shorts/6YHdfctSgGk?feature=share

Executive Summary

The clock pendulum project was intended to learn the fundamentals of Computer Aided Design (CAD) using Autodesk Fusion and numerous fabrication skills, such as laser cutting and drilling. These skills are then used to fabricate and assemble a pendulum clock that would rotate an escapement wheel clockwise and oscillate a pendulum beneath it. The main objective of the analysis for this project was the clock's natural frequency, in which two different methods were tested: a point-mass pendulum analysis and a rigid body pendulum analysis. From the results, the measured time for one revolution of the escapement wheel was 10.07 s. The point-mass analysis gave a calculated time of one revolution to be approximately 8.01 s, which resulted in an error of 20.42% between the expected and measured value. The rigid body analysis, on the other hand, had a calculated time for one revolution of the wheel to be 9.72 s, which resulted in a 3.41% error in the results. When comparing both methods of analysis, it was concluded that a rigid body assumption gave a much more accurate result when calculating the natural frequency of the pendulum clock; thus, performing a rigid body analysis would be more suitable than assuming all mass is concentrated at one point of the pendulum for analysis.

Analysis

The pendulum design of the clock was designed using the CAD software known as Autodesk Fusion, or Fusion 360. Below are screenshots of the 3D design in Fusion from an isometric view.

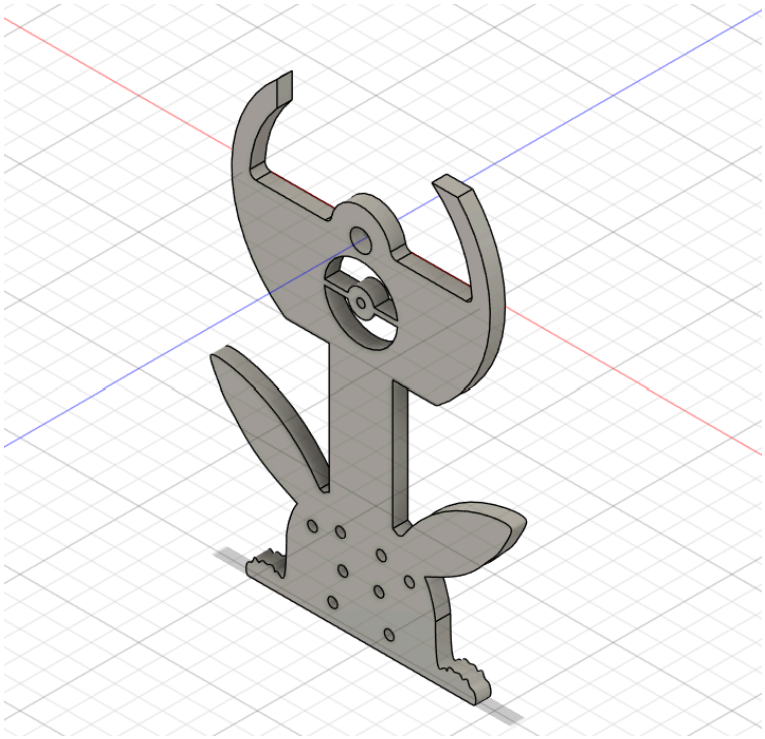


Fig 1. Isometric view of pendulum design

Additionally, the Fusion software also automatically calculates other values for the pendulum based on certain reference points within the coordinate system and the type of material. Using the Properties tab in the software, we can gather that the density (p) of the pendulum is 1.188 g/cm^3 and the center of mass for the acrylic (L_a) is approximately 6.473 cm in the -y-direction from the origin of the pendulum (the origin of the pendulum is the center of the hole where the small shaft is inserted).

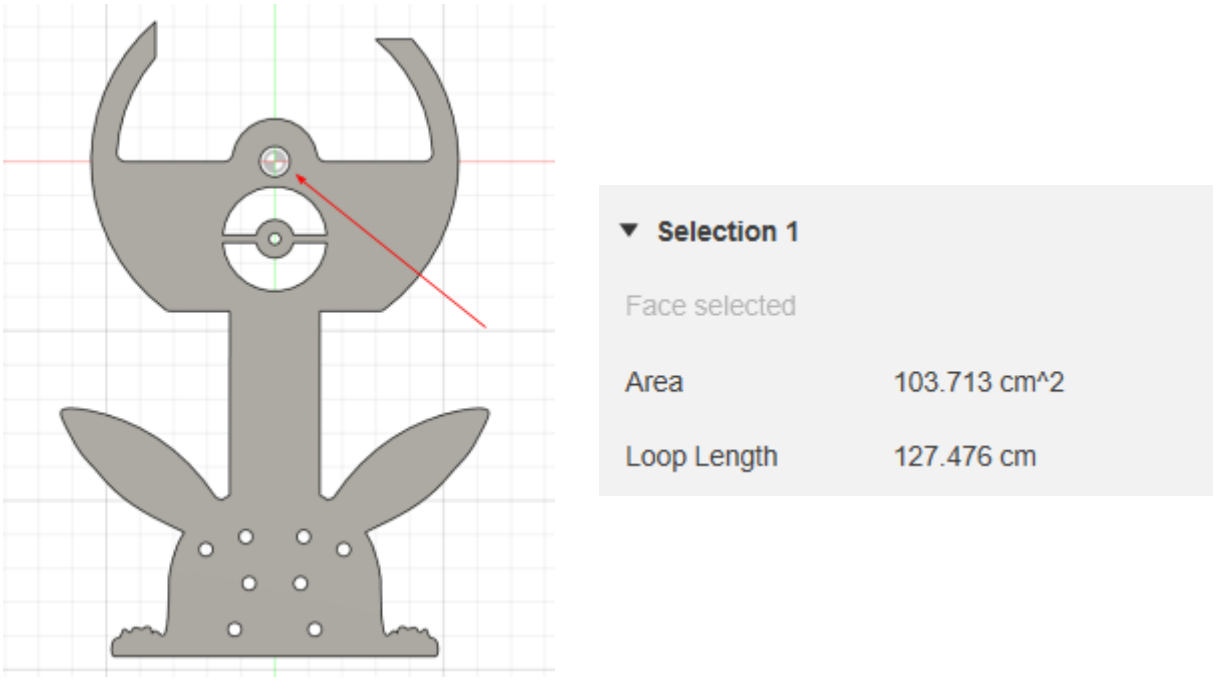


Fig 2. Front view with location of origin (indicated by red arrow) and corresponding Area value

Additionally, the area (A) of the pendulum is found to be 103.713 cm^2 and the thickness (t) of the pendulum is .635 cm. Using these values, the calculated volume (V) of the pendulum is:

$$V = A * t = 65.858\, cm^3 \quad (1)$$

And the calculated mass (M_{calc}) of the pendulum without weights is

$$M_{calc} = \rho * V = 78.24g \quad (2)$$

To find the total mass of the pendulum with the weights, the equation is

$$M_t = M_{calc} + M_{bolts} * N_{bolts} \quad (3)$$

M_{bolts} is the mass of the bolts, and N_{bolts} is the number of bolts. Taking M_{bolts} to be 4.0g and N_{bolts} to be 8, we get an M_t of 110.24g.

For the analysis of the clock timing, two methods of analysis were used: a point-mass analysis and a rigid body analysis.

Point-Mass Assumption

For calculating the timing of the pendulum clock by treating it as a point-mass system, the following assumptions must be made:

1) Negligible friction
2) Small angle of oscillation
3) The mass of the pendulum is concentrated at one point
4) Pendulum rope does not stretch
5) Pendulum frequency unaffected by contact with the escapement wheel

To calculate the natural frequency of the pendulum using a point-mass analysis, the following equation is used:

$$\omega_{point} = \sqrt{\frac{g}{L_{com}}} \qquad [1] (4)$$

Here, the gravitational constant (g) is assumed to be 9.8 m/s^2, and L_{com} is the effective length of the pendulum. To find L_{com}, the additional weight from the bolts must be considered to properly estimate the pendulum's center of mass. Below is a screenshot of the length of each bolt from the origin and a table with all values of the length to the origin of each bolt in meters.

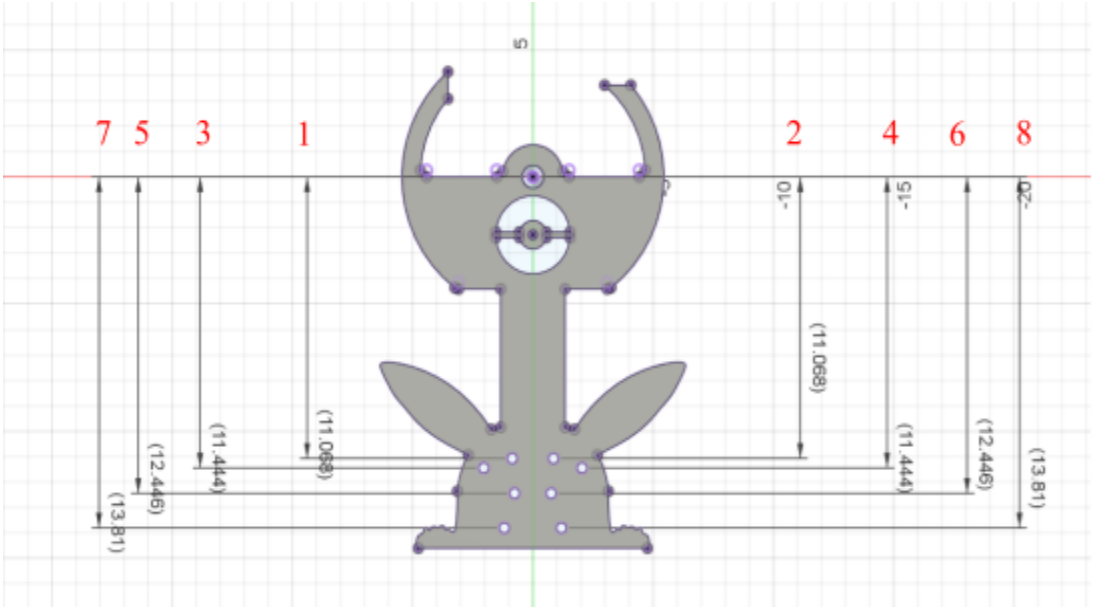


Fig 3. Vertical length of each bolt from the origin

Table 1. Length of Each Bolt from the Origin Point of the Pendulum

Bolt Number	**Length from Origin (m)**
1	.11068
2	.11068
3	.11444
4	.11444
5	.12446
6	.12446
7	.1381

8	.1381

To find the length of the center of mass of the pendulum, the following equation was used:

$$L_{com} = \frac{M_{calc}{}^{*}L_a + M_{bolt}{}^{*}(\sum_{n=1}^{8} L_{bolt,n})}{M_t} = .0813\ m \qquad (5)$$

With these results, we are now able to find the natural frequency of the pendulum with the point-mass assumption method. Plugging in g and L_{com} into equation (4), the equation finally gives us

$$\omega_{point} = 10.98\ \frac{rad}{s} \Rightarrow f = \frac{\omega}{2\pi} = 1.75\ Hz \qquad (6)$$

Rigid Body Assumption

For a rigid body analysis, the following assumptions are made when calculating the natural frequency:

1) Negligible friction
2) Small angle of oscillation
3) Pendulum frequency unaffected by contact with the escapement wheel

The following equation is used to calculate the natural frequency using a rigid body analysis:

$$\omega_{rigid} = \sqrt{\frac{M_t{}^{*}g^{*}L_{com}}{I}} \qquad [1]\ (7)$$

To utilize this equation, the total moment of inertia (I_{tot}) would need to be found. The total moment of inertia for the whole pendulum is

$$I_{total} = I_a + \sum_{n=1}^{8} I_{bolt,n} \qquad (8)$$

With I_a being the moment of inertia of the acrylic pendulum and $I_{bolt,n}$ being the moment of inertia for each bolt. To find each moment of inertia for the bolt, the following equation is used:

$$I_{bolt,n} = mr_n^2 \qquad [1]\ (9)$$

The moment of inertia for the pendulum can be found on Fusion 360 and looking up the value for "I_{ZZ}" in the Properties tab.

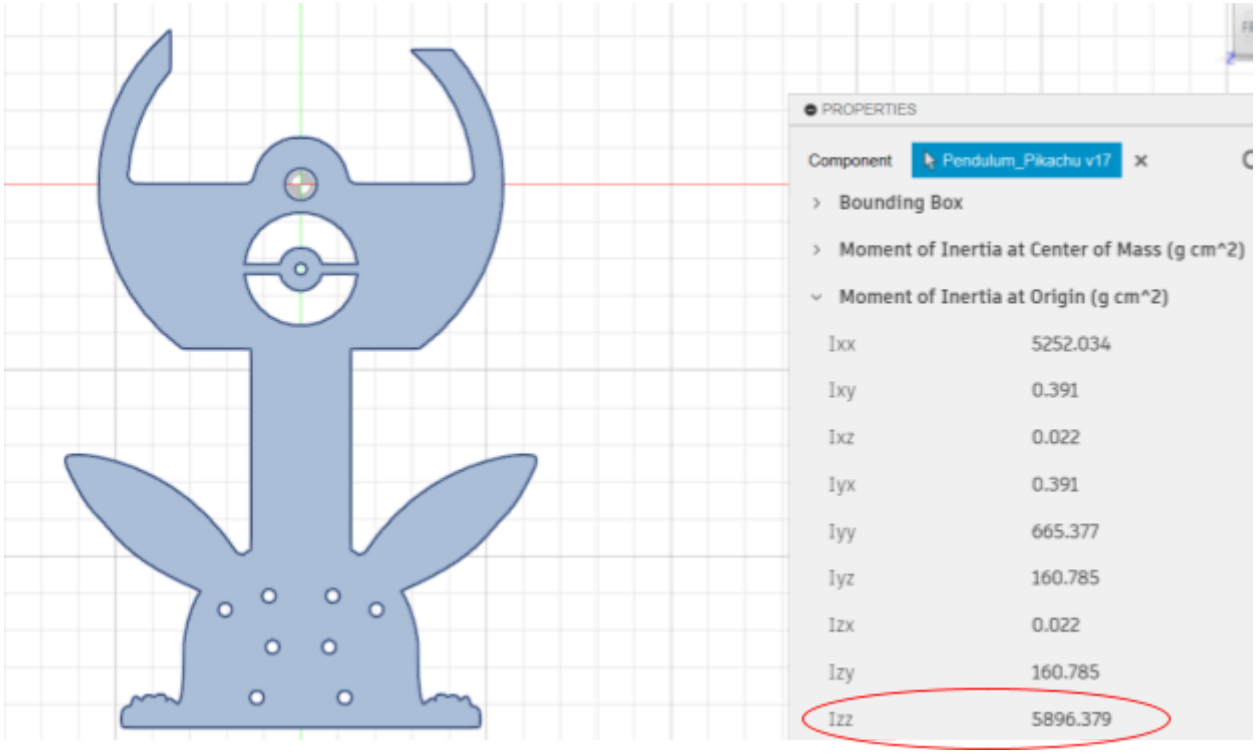


Fig 4. Pendulum with its Moment of Inertia Value (I_{ZZ})

Additionally, all moments of inertia from each bolt must be taken into account as well. Below is a screenshot of the distance of each bolt from the origin of the pendulum and a table with each bolt's calculated moment of inertia.

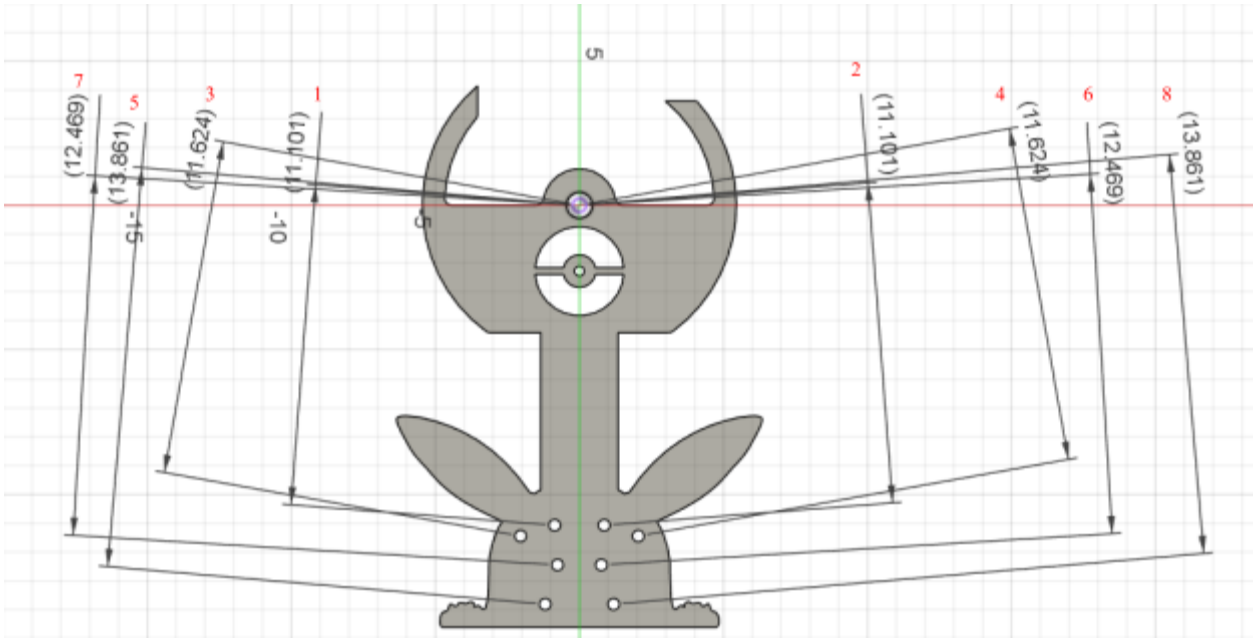


Fig 5. Distance from the origin for each bolt

Table 2. Distance of Each Bolt from the Origin of the Pendulum

Bolt Number	**Moment of Inertia (g*cm^2)**
1	492.93
2	492.93
3	540.47
4	540.47
5	621.90
6	621.90
7	768.51

8	768.51

With these values, the total moment of inertia I_{tot} is calculated to be

$$I_{tot} = 10744.002\ g * cm^2 \qquad (10)$$

The natural frequency of the pendulum using a rigid body analysis was calculated to be

$$\omega_{rigid} = 9.04\ \frac{rad}{s} \Rightarrow f = 1.44\ Hz \qquad (11)$$

Experimental Results

Using a calibrated scale, the pendulum design without the nuts and bolts was weighed to find the percent error of the calculated value. The actual mass of the pendulum (M_{Act}) was found to be 64.0 g. Using a simple percent error calculation:

$$M_{\%\ error} = \frac{M_{calc} - M_{act}}{M_{act}} * 100 = 22.25\ \% \qquad (12)$$

To calculate the percent error of this length from the center of mass, intermediate verification was done by balancing the physical pendulum, with the bolts attached, on one finger and measuring the distance between the origin of the pendulum and where that finger was placed. A precise caliper was used to get an estimated length from the center of mass of .0850 m. Using the percent error formula, the following percentage was calculated:

$$L_{com,\%\ error} = \frac{L_{com,calc} - L_{com,est}}{L_{com,est}} * 100 = 4.36\ \% \qquad (13)$$

To compare both methods for discussion, the period of one revolution for the escapement wheel was found (T_{meas}). The period was calculated by using the finished clock pendulum and measuring the time it took for one full revolution of the escapement wheel to be completed. The

result was that it took 10.07 s for the wheel to complete one revolution. To find the calculated values from each method, the reciprocal of the natural frequency (Hz) was taken to be equal to the period of oscillation for the pendulum. This period was then multiplied by the number of teeth the wheel had (14 teeth in total) to get a calculated time for one revolution of the escapement wheel. The results are as follows:

Table 3. Calculated Values from Both Methods

	Point-Mass	**Rigid Body**
Natural Frequency ω (rad/s)	10.98	9.04
Natural Frequency ω (Hz)	1.75	1.44
Period of Oscillation T (s)	.572	.694
Time of One Revolution (s)	8.013	9.727

The percent error in clock timing for both methods was calculated using the calculated times for one revolution of the escapement.

$$t_{\%\ error} = \frac{t_{calc} - t_{meas}}{t_{meas}} * 100 \qquad (14)$$

$$\Rightarrow t_{\%\ error,\ point} = 20.42\ \% \qquad (15)$$

$$\Rightarrow t_{\%\ error,\ rigid} = 3.41\ \% \qquad (16)$$

Refer to Appendix A and B for an Excel spreadsheet view of the evaluation of the calculated and experimental results of the pendulum clock.

Discussion

From the calculated theoretical values and experimental values obtained, one can make logical conclusions based on the method of calculating the natural frequency of the pendulum clock. To start, the calculated mass of the acrylic pendulum had a resulting percent error of 22.25% from the measured mass of the pendulum. The reason for this discrepancy in mass calculation is mainly due to the loss of mass from fabricating the pendulum. To get the desired design of the pendulum, a Lasercamm was used to laser cut the design from a stock rectangular piece of acrylic. The process of laser cutting involves melting mass off the metal using a laser until the laser has completely cut out the part to its desired length. The process being strictly heat-induced alludes to the possibility that some of the material was lost during the process of laser cutting, thus explaining the lower measured result compared to the calculated value.

The length from the center of mass was also analyzed for error with intermediate verification. The calculated length to the center of mass compared to the measured length from the center of mass had a resulting percent error of 4.36%. The reason for this discrepancy in the data is mainly due to human error and estimation. The measured length to the center of mass was verified by balancing the pendulum on one finger and by measuring the distance between this finger from the origin of the pendulum using a set of calipers. This method of verifying a calculated value is subject to error as it relies on the accuracy of the human eye to find the exact distance from the center of the mass of the object.

Lastly, the natural frequency calculations from both a point-mass assumption and a rigid body assumption were compared to the experimental natural frequency of the pendulum to evaluate the accuracy of each method. The following methods were utilized in this experiment to show how accurate a method is, depending on the number of assumptions the method uses to find a result. From the experimental results, it is observed that the rigid body assumption resulted in a lower percent error compared to a point-mass assumption. The reason for this result is mainly due to the assumptions each method makes. A point-mass assumption simplifies a given model by assuming that all the mass of an object is concentrated at one point, thus neglecting the object's shape and size. A rigid body assumption, however, takes into account the shape and size of the object and focuses on the rotational motion of the object. By removing the assumption that all the mass is focused at one point during the motion, the rigid body assumption more closely reflects the real world. While a rigid body assumption does more accurately represent the natural frequency compared to a point-mass analysis, there are still errors present due to circumstances not accounted for in the method. For example, both methods assume that friction is negligible, which will surely skew the results, as there is friction present where the escapement wheel makes contact with the arms of the pendulum design. Additionally, each method assumes that contact with the escapement wheel will not affect the frequency of the pendulum oscillation. As a result of this assumption, both methods calculated the period of oscillation for the pendulum to be much quicker compared to the measured period.

References

[1] OpenStax. (n.d.) *15.5: Pendulums.* In *University Physics I - Mechanics, Sound, Oscillations, and Waves.* Physics LibreTexts. Retrieved April 20, 2025, phys.libretexts.org/Bookshelves/University_Physics/University_Physics_(OpenStax)/Book%3A_University_Physics_I_-_Mechanics_Sound_Oscillations_and_Waves_(OpenStax)/15%3A_Oscillations/15.05%3A_Pendulums

Appendix

A. Pendulum Timing Analysis (Calculated Values)

Pendulum Timing Analysis

Name: Joshua Conge
Section: B02

Variable Description	Variable Name	Values/Equations	Units	Comments
Acrylic Pendulum Specifications				
Area	A	103.713	cm^2	From Fusion360
Thickness	t	0.635	cm	
Volume	Vol	65.857755	cm^3	
Density	p	1.188	g/cm^3	
Calculated Mass of Acrylic	M_Calc	78.23901294	g	Volume*density
Actual Mass of Acrylic	M_Act	64	g	Measured physically
Length to Center of Mass of Acrylic	La	6.473	cm	acrylic only
Percent Error in Acrylic Mass Calculation	M_Error	22.24845772	%	
Calculate Total Mass of Pendulum				
Mass of One Bolt with Two Nuts	Mb	4	g	Given
Number of Bolts with Two Nuts	Nb	8		
Total Mass of Pendulum with Nuts and Bolts	Mt	110.2390129	g	calculated
Calculate Center of Mass of Pendulum with Bolts				
Length to Center of Mass of Bolt 1	L_bolt1	0.11068	meters	distance from pivot point to bolt
Length to Center of Mass of Bolt 2	L_bolt2	0.11068	meters	distance from pivot point to bolt
Length to Center of Mass of Bolt 3	L_bolt3	0.11444	meters	distance from pivot point to bolt
Length to Center of Mass of Bolt 4	L_bolt4	0.11444	meters	distance from pivot point to bolt
Length to Center of Mass of Bolt 5	L_bolt5	0.12446	meters	distance from pivot point to bolt
Length to Center of Mass of Bolt 6	L_bolt6	0.12446	meters	distance from pivot point to bolt

Length to Center of Mass of Bolt 7	L_bolt7	0.1381	meters	distance from pivot point to bolt
Length to Center of Mass of Bolt 8	L_bolt8	0.1381	meters	distance from pivot point to bolt
Length to Center of Mass **in Meters**	Lcom_meter	0.08133101947	**meters**	calculate with acrylic and bolts
Estimated Center of Mass of Pendulum with Nuts and Bolts	Lcom_est	0.085042	meters	intermediate verification with acrylic and bolts
Percent Error in Pendulum Nuts and Bolts Lcom Estimate	Lcom_error	4.363703266	%	
Calculate Natural Frequency and Timing using Point Mass Assumption				
Gravitational Constant	g	9.8	m/s^2	
Natural Frequency in radians/sec	nat_freq_rad_sec	10.97703199	rad/s	
Natural Frequency in Hz	nat_freq_hz	1.747048901	Hz	
Period of Oscillation	period	0.5723938232	s	
Number of Teeth on Escapement Wheel	nteeth	14		
Calculated Time of One Revolution of Escapement Wheel	time_calc	8.013513525	s	
Measured Time of One Revolution of Escapement Wheel	time_meas	10.07	s	
Percent Error in Clock Timing	time_error	20.42191137	%	
Calculate Natural Frequency and Timing using Rigid Body Assumption				
Moment of Inertia of Pendulum	I_a	5896.379	g*cm^2	
Moment of Inertia of Bolt 1	I_bolt1	492.928804	g*cm^2	
Moment of Inertia of Bolt 2	I_bolt2	492.928804	g*cm^2	symmetric with I_1
Moment of Inertia of Bolt 3	I_bolt3	540.469504	g*cm^2	
Moment of Inertia of Bolt 4	I_bolt4	540.469504	g*cm^2	symmetric with I_3
Moment of Inertia of Bolt 5	I_bolt5	621.903844	g*cm^2	
Moment of Inertia of Bolt 6	I_bolt6	621.903844	g*cm^2	symmetric with I_5
Moment of Inertia of Bolt 7	I_bolt7	768.509284	g*cm^2	

Moment of Inertia of Bolt 8	I_bolt8	768.509284	g*cm^2	symmetric with I_7
Total Moment of Inertia	I_total	10744.00187	g*cm^2	
Natural Frequency in radians/sec	rb_nat_freq_rad_sec	9.043275595	rad/s	
Natural Frequency in Hz	rb_nat_freq_hz	1.439282013	Hz	
Period of Oscillation	rb_period	0.6947908687	s	
Calculated Time of One Revolution of Escapement Wheel	rb_time_calc	9.727072162	s	
Percent Error in Clock Timing	rb_time_error	3.405440294	%	

B. Pendulum Timing Analysis (Equations w/ named variables)

Pendulum Timing Analysis

Name: Joshua Conge				
Section: B02				
Variable Description	**Variable Name**	**Values/Equations**	**Units**	**Comments**
Acrylic Pendulum Specifications				
Area	A	103.713	cm^2	From Fusion360
Thickness	t	0.635	cm	
Volume	Vol	=A*t	cm^3	
Density	p	1.188	g/cm^3	
Calculated Mass of Acrylic	M_Calc	=Vol*p	g	Volume*density
Actual Mass of Acrylic	M_Act	64	g	Measured physically
Length to Center of Mass of Acrylic	La	6.473	cm	acrylic only
Percent Error in Acrylic Mass Calculation	M_Error	=ABS(M_Calc-M_Act)/M_Ac	%	

		t * 100		
Calculate Total Mass of Pendulum				
Mass of One Bolt with Two Nuts	Mb	4	g	Given
Number of Bolts with Two Nuts	Nb	8		
Total Mass of Pendulum with Nuts and Bolts	Mt	=M_Calc + Mb*Nb	g	calculated
Calculate Center of Mass of Pendulum with Bolts				
Length to Center of Mass of Bolt 1	L_bolt1	0.11068	meters	distance from pivot point to bolt
Length to Center of Mass of Bolt 2	L_bolt2	=L_bolt1	meters	distance from pivot point to bolt
Length to Center of Mass of Bolt 3	L_bolt3	0.11444	meters	distance from pivot point to bolt
Length to Center of Mass of Bolt 4	L_bolt4	=L_bolt3	meters	distance from pivot point to bolt
Length to Center of Mass of Bolt 5	L_bolt5	0.12446	meters	distance from pivot point to bolt
Length to Center of Mass of Bolt 6	L_bolt6	=L_bolt5	meters	distance from pivot point to bolt
Length to Center of Mass of Bolt 7	L_bolt7	0.1381	meters	distance from pivot point to bolt
Length to Center of Mass of Bolt 8	L_bolt8	=L_bolt7	meters	distance from pivot point to bolt
Length to Center of Mass **in Meters**	Lcom_meter	=(M_Calc*(La/100)+Mb*(L_bolt1+L_bolt2+L_bolt3+L_bolt4+L_bolt5+L_bolt6+L_bolt7+L_bolt8))/Mt	**meters**	calculate with acrylic and bolts
Estimated Center of Mass of Pendulum with Nuts and Bolts	Lcom_est	0.085042	meters	intermediate verification with acrylic and bolts
Percent Error in Pendulum Nuts and Bolts Lcom Estimate	Lcom_error	=ABS(Lcom_meter-Lcom_est)/Lcom_est * 100	%	
Calculate Natural Frequency and Timing using Point Mass Assumption				
Gravitational Constant	g	9.8	m/s^2	
Natural Frequency in radians/sec	nat_freq_rad_sec	=SQRT(g/Lcom_meter)	rad/s	
Natural Frequency in Hz	nat_freq_hz	=nat_freq_rad_sec/(2*PI())	Hz	
Period of Oscillation	period	=1/nat_freq_h	s	

		z		
Number of Teeth on Escapement Wheel	nteeth	14		
Calculated Time of One Revolution of Escapement Wheel	time_calc	=period*14	s	
Measured Time of One Revolution of Escapement Wheel	time_meas	10.07	s	
Percent Error in Clock Timing	time_error	=ABS(time_calc-time_meas)/time_meas * 100	%	
Calculate Natural Frequency and Timing using Rigid Body Assumption				
Moment of Inertia of Pendulum	I_a	5896.379	g*cm^2	
Moment of Inertia of Bolt 1	I_bolt1	=Mb*(11.101^2)	g*cm^2	
Moment of Inertia of Bolt 2	I_bolt2	=I_bolt1	g*cm^2	symmetric with I_1
Moment of Inertia of Bolt 3	I_bolt3	=Mb*(11.624^2)	g*cm^2	
Moment of Inertia of Bolt 4	I_bolt4	=I_bolt3	g*cm^2	symmetric with I_3
Moment of Inertia of Bolt 5	I_bolt5	=Mb*(12.469^2)	g*cm^2	
Moment of Inertia of Bolt 6	I_bolt6	=I_bolt5	g*cm^2	symmetric with I_5
Moment of Inertia of Bolt 7	I_bolt7	=Mb*(13.861^2)	g*cm^2	
Moment of Inertia of Bolt 8	I_bolt8	=I_bolt7	g*cm^2	symmetric with I_7
Total Moment of Inertia	I_total	=I_a+I_bolt1+I_bolt2+I_bolt3+I_bolt4+I_bolt5+I_bolt6+I_bolt7+I_bolt8	g*cm^2	
Natural Frequency in radians/sec	rb_nat_freq_rad_sec	=SQRT(Mt*g*Lcom_meter/(I_total/100^2))	rad/s	
Natural Frequency in Hz	rb_nat_freq_hz	=rb_nat_freq_rad_sec/(2*PI())	Hz	
Period of Oscillation	rb_period	=1/rb_nat_freq_hz	s	
Calculated Time of One Revolution of Escapement Wheel	rb_time_calc	=14 * rb_period	s	

Percent Error in Clock Timing	rb_time_error	=ABS(rb_time_calc-time_meas)/time_meas * 100	%	